

GCRew_Flume_TidalSurvey_DIC_QAQC_202402

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##Setup - Change things here & write any notes

```
#identify section  
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE  
Date_Run = "08/25/2023" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = " The standard curve for this run was checked manually on the instrument. " #any notes for this run  
samples_pattern <- "Flume"  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_" #what did you name your check standards?  
crm_name = "CRM|crm" #what did you name your CRMS?  
  
library(here)
```

Warning: package 'here' was built under R version 4.3.3

here() starts at S:/Biogeochemistry/People/Steph Wilson/Github/serc-lateral-flux

```
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
  #raw_file_name = "Raw Data/TOCTN_COMPASS_Synoptic_DIC_202305.txt"  
raw_file_name = c(here("Tidal_Surveys/Raw_Data/202402", "TOCT_Flume_TidalSurvey_Feb2024_Neap_DIC.txt"),  
  here("Tidal_Surveys/Raw_Data/202402", "TOCT_Flume_TidalSurvey_Feb2024_Spring_DIC.txt"))  
  
#file path and name for raw all peaks file  
  #raw_allpeaks_name = "Raw Data/TOCTN_COMPASS_Synoptic_DIC_202305_allpeaks.txt"  
  
#file path and name of processed data file  
  #processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DIC_202308.csv"  
processed_file_name = here("Tidal_Surveys/Processed_Data/202402", "Flume_TidalSurvey_DIC_Processed_Data.csv")  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = here("Tidal_Surveys/Raw_Data/202402", "Tidal_Survey_202402_Metadata.csv")
```

```
#qaqc log file path for this year  
# Log_path = "Raw Data/COMPASS_Synoptic_DIC_QAQClog_2023.csv"
```

```
##Set Up Code
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## New names:
```

```
## * ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,  
## e.g.:
```

```
## dat <- vroom(...)
```

```
## problems(dat)
```

```
## # A tibble: 6 x 3
```

```
## sample_name          ic_raw run_datetime
```

```
## <chr>                <dbl> <chr>
```

```
## 1 20240208_GCReW_Flume_0900 10 2/12/2024 12:23:39 PM
```

```
## 2 20240208_GCReW_Flume_1000 9.72 2/12/2024 12:35:53 PM
```

```
## 3 20240208_GCReW_Flume_1100 6.94 2/12/2024 12:47:57 PM
```

```
## 4 20240208_GCReW_Flume_1200 9.50 2/12/2024 1:00:08 PM
```

```
## 5 20240208_GCReW_Flume_1300 9.46 2/12/2024 1:12:16 PM
```

```
## 6 20240208_GCReW_Flume_1400 9.90 2/12/2024 1:24:22 PM
```

Assessing Standard Curves - this was done manually on the TOC, not included here

CRMs Check: CRM Batch 221-0866

```
## Assess the CRMs
```

```
## New names:
```

```
## New names:
```

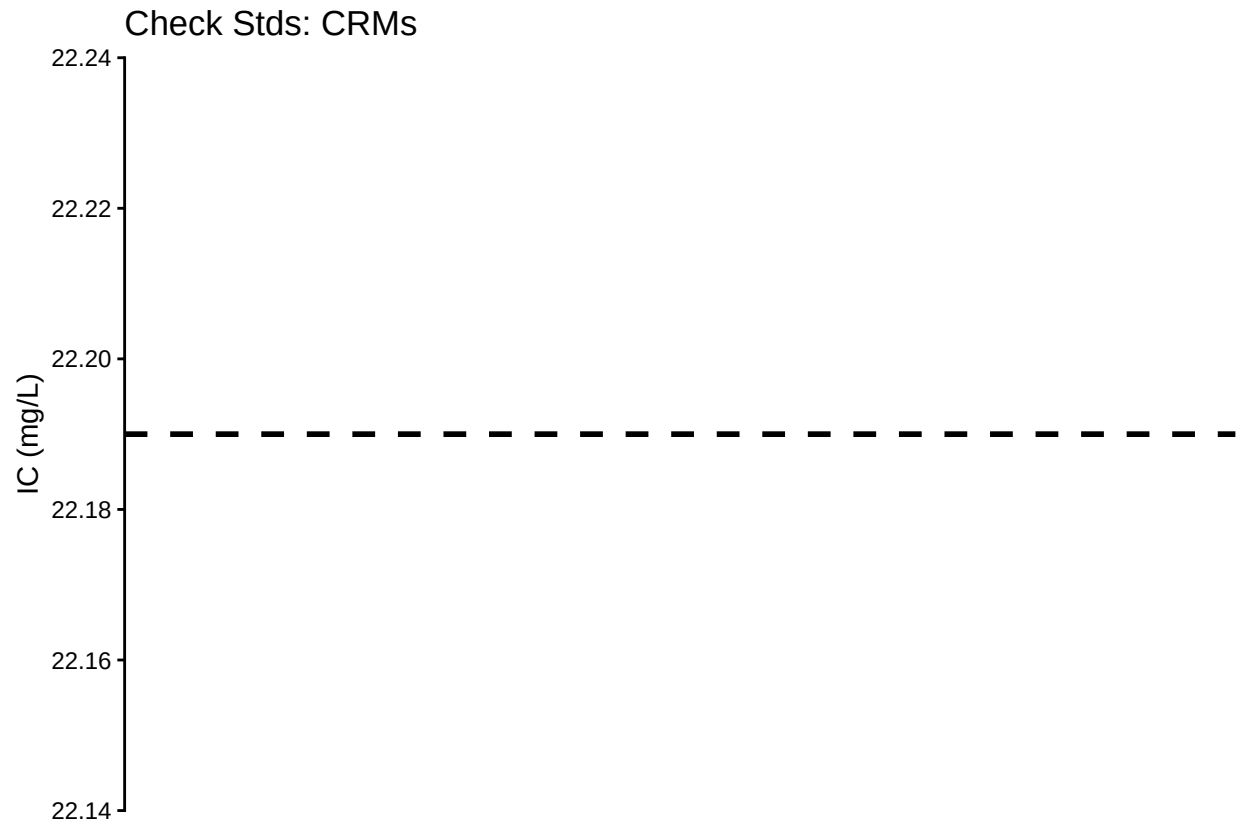
```
## * ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,  
## e.g.:
```

```
## dat <- vroom(...)
```

```
## problems(dat)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the  
## data's fill values.
```



```
## [1] NA
```

```
## Run mean = NaN
```

```
## Expected = 22.19
```

Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
```

```
## New names:
```

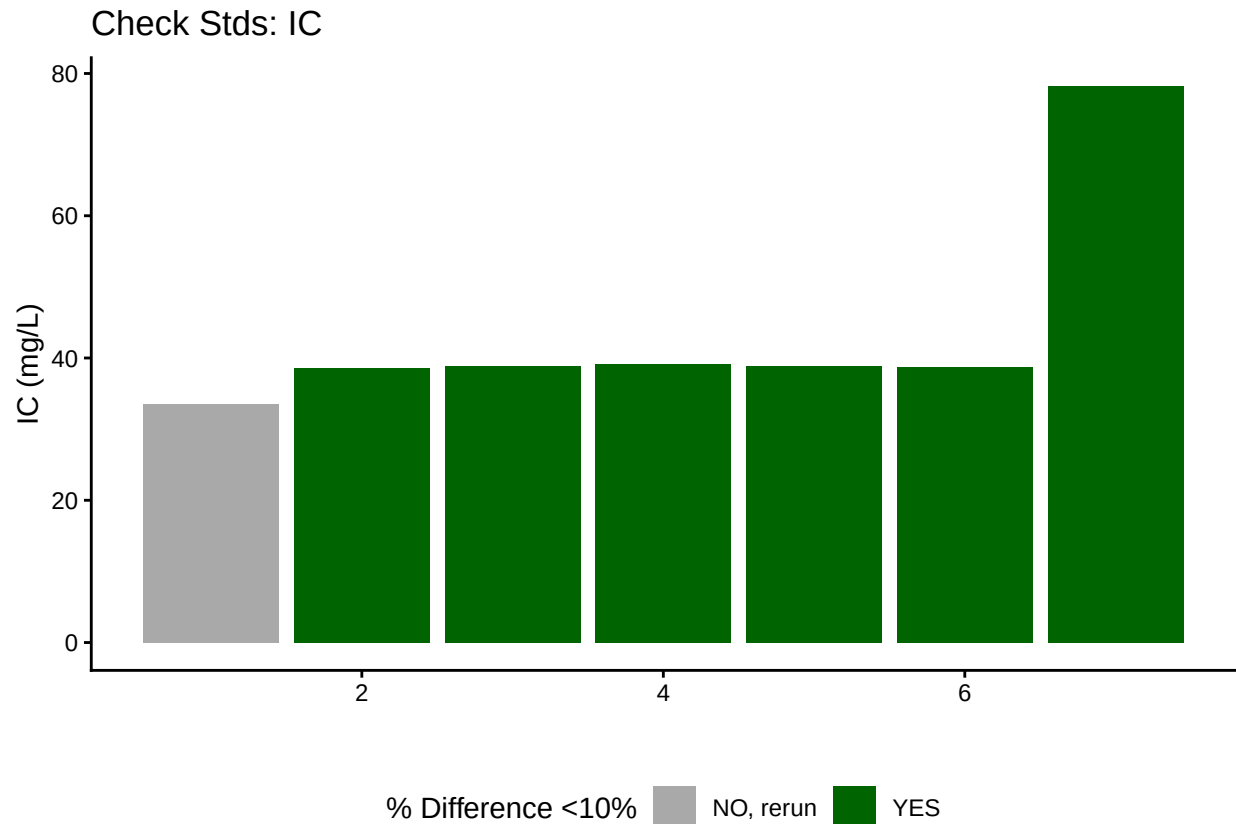
```
## * ' ' -> '...14'
```

```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
```

```
## e.g.:
```

```
## dat <- vroom(...)
```

```
## problems(dat)
```



```
## [1] ">60% of IC Check Standards are within range of expected concentration"
```

Assess Blanks

```
## Assess Blanks
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

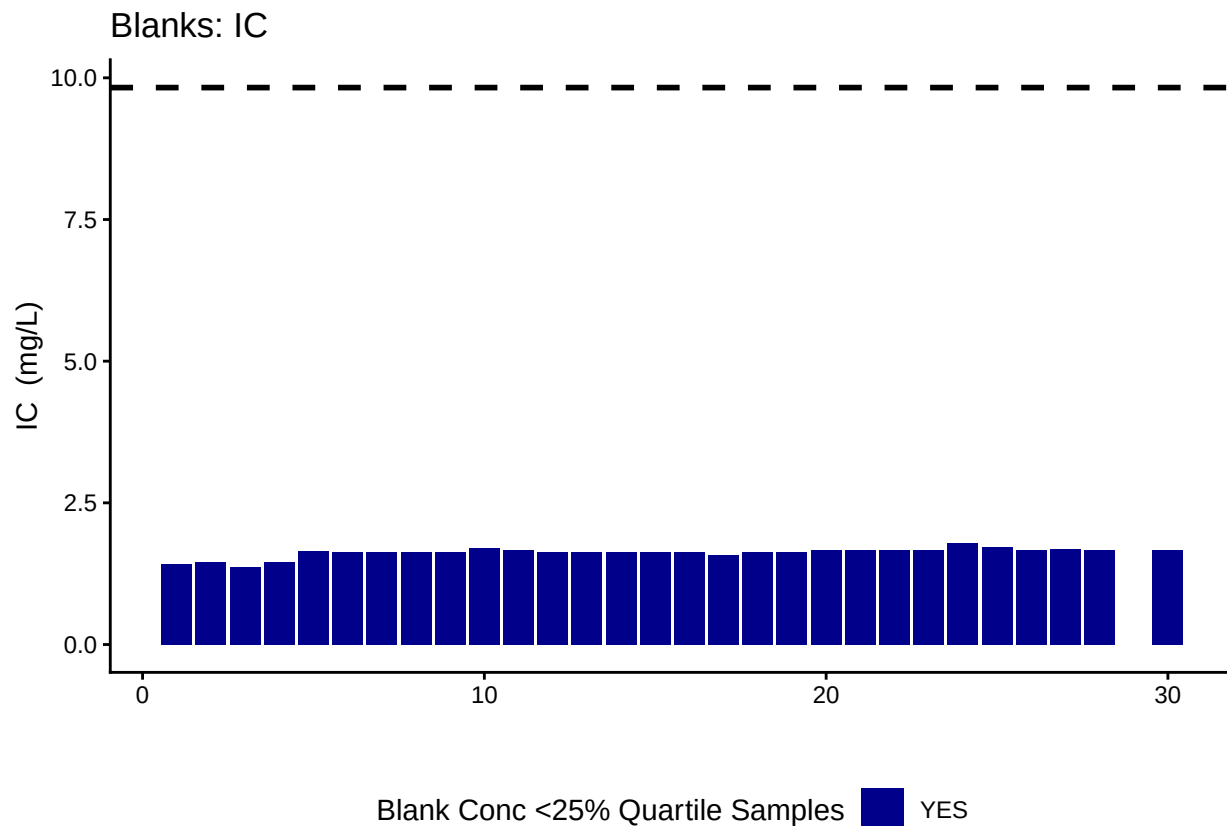
```
## Warning: One or more parsing issues, call 'problems()' on your data frame for details,
```

```
## e.g.:
```

```
##   dat <- vroom(...)
```

```
##   problems(dat)
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 1.625345
```

Assess Duplicates - none on this run

```
“{#r Check Duplicates, echo=FALSE}
```

```
cat(“Assess Duplicates”)
```

```
#Take a look at the raw data #head(dat_raw)
```

```
#pull out any rows that have “dup” in the sample_name column dups <- dat_raw %>%
select(!c(ic_flag)) %>% filter(str_detect(sample_name, “dup”)) #have to change this to match data
```

```
#create a new dataframe and remove dups from sample dataframe dat_raw2 <- dat_raw %>%
filter(!str_detect(sample_name, “dup”))
```

```
#remove the dup from these IDs so we will have duplicate sample names dupssample_name <-
gsub(“dup”, “”, as.character(dupssample_name)) dups <- dups[, -c(3)] #remove the run date time
for colnames(dups) <- c(“sample_name”, “ic_raw_dup”)
```

```
#merge with the dataframe so we have a column for the conc and the dup QAdups <- merge(dat_raw2,
dups)
```

```
#create a dataframe to compare npoc dups df2 <- as.data.frame(QAdups)df2dups <- QAdups$ic_raw_dup
```

```
#calculate the cv of the duplicates df2sds <- apply(df2, 1, sd)df2mean <- apply(df2, 1, mean)
```

```

QAdupsic $_{dups_{cv}} < -(df2sds/df2mean) * 100$ QAdupsic_dups_cv_flag <- ifelse(QAdups$ic_dups_cv <10,
'YES', 'NO, rerun')

#Put all the dups together and create row numbers for plotting QAdups <- QAdups %>% mutate(row_num
= row_number()) #head(QAdups)

#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this C_dups
<- ggplot(data =QAdups, aes(x =row_num, y =ic_dups_cv, fill=ic_dups_cv_flag)) + geom_bar(stat
= 'identity') + theme_classic() + labs(x= " ", y="CV of IC Duplicates") + scale_fill_manual(values =
c("YES" = "darkgreen", "NO, rerun" = "red")) + theme(legend.position="none") + geom_hline(yintercept=10,
linetype="dashed", color = "black", linewidth=1) + guides(fill=guide_legend(title="CV Between Dups
<10%"))

C_dups

#calculate the percent of check standards that are within the range based on the flag c_dups_percent <-
(sum(QAdups$ic_dups_cv_flag == "YES")/nrow(QAdups))*100

#report out if the dups are within range ifelse(c_dups_percent >= chks_flag, ">60% of Carbon Duplicates
have a CV <10%", "<60% of Carbon Duplicates have a CV <10% - REASSESS")

#write out a flag to the sample dataframe if more than 60% of the dups have CVs out of range if
(c_dups_percent <= chks_flag) { dat_rawic_flag <- ifelse(dat_rawic_flag != "", paste0(dat_raw$ic_flag, ";
IC dups out of range"), "IC dups out of range" ) }

```

```
## Sample Flagging - Are samples Within the range of the curve?
```

Sample Flagging

```

<!-- -->

## Visualize Data by Plot - in 2023 SWH Upland = UPCON and Swamp = UP

```

Visualize Data

Warning in rbind(c("Flume", "20240208", "SW", "0900"), c("Flume", "20240208", :

number of columns of result is not a multiple of vector length (arg 1)

```

<!-- -->

## Convert data from mg/L to uMoles/L

## Match metadata & merge info

```

Merge with Metadata

Survey__Type Site Date Time__EDT Time__EST Sample__Type

1 Tidal Survey Flume 2/8/2024 9:00 10:00 SW

2 Tidal Survey Flume 2/8/2024 10:00 11:00 SW

3 Tidal Survey Flume 2/8/2024 11:00 12:00 SW

4 Tidal Survey Flume 2/8/2024 12:00 13:00 SW

5 Tidal Survey Flume 2/8/2024 13:00 14:00 SW

6 Tidal Survey Flume 2/8/2024 14:00 15:00 SW

Sample__ID DO__mgL pH Salinity__ppt Temp DIC DICex SO4__H2S NUTR
DOC

1 Flume__20240208__SW__0900 NA NA x x x x x

2 Flume__20240208__SW__1000 NA NA x x x x x

3 Flume__20240208__SW__1100 NA NA x x x x x

4 Flume__20240208__SW__1200 NA NA x x x x x

5 Flume__20240208__SW__1300 NA NA x x x x x

6 Flume__20240208__SW__1400 NA NA x x x x x

GHGs TA Radium__Vol__L Notes

1 x x 5 February: Neap Tide

2 x x 5 February: Neap Tide

3 x x 5 February: Neap Tide

4 x x 5 February: Neap Tide

5 x x 5 February: Neap Tide

6 x x 5 February: Neap Tide

Some sample IDs are missing from metadata.

- [1] “Flume_20240208_SW_0900” “Flume_20240208_SW_1000”
- [3] “Flume_20240208_SW_1100” “Flume_20240208_SW_1200”
- [5] “Flume_20240208_SW_1300” “Flume_20240208_SW_1400”
- [7] “Flume_20240208_SW_1500” “Flume_20240208_SW_1600”
- [9] “Flume_20240208_SW_1600_isco” “Flume_20240208_SW_1700”
- [11] “Flume_20240208_SW_1700_isco” “Flume_20240208_SW_1800”
- [13] “Flume_20240208_SW_1800_isco” “Flume_20240208_SW_1900”
- [15] “Flume_20240208_SW_1900_isco” “Flume_20240208_SW_2000”
- [17] “Flume_20240208_SW_2000_isco” “Flume_20240208_SW_2100”
- [19] “Flume_20240208_SW_2100_isco” “Flume_20240208_SW_2200_isco”
- [21] “Flume_20240208_SW_2300_isco” “Flume_20240208_SW_2400_isco”
- [23] “Flume_20240209_SW_0000_isco” “Flume_20240209_SW_0100_isco”
- [25] “Flume_20240209_SW_0200_isco” “Flume_20240209_SW_0300_isco”
- [27] “Flume_20240209_SW_0400_isco” “Flume_20240208_GW_1”
- [29] “Flume_20240208_GW_2” “Flume_20240208_GW_3”
- [31] “Flume_20240208_GW_4” “Flume_20240208_GW_5”
- [33] “Flume_20240209_FW_1” “Flume_20240209_FW_2”
- [35] “Flume_20240215_SW_0800” “Flume_20240215_SW_0900”
- [37] “Flume_20240215_SW_1000” “Flume_20230215_SW_1100”
- [39] “Flume_20230215_SW_1200” “Flume_20230215_SW_1300”
- [41] “Flume_20230215_SW_1400” “Flume_20230215_SW_1500”
- [43] “Flume_20230215_SW_1600” “Flume_20230215_SW_1700”
- [45] “Flume_20230215_SW_1800” “Flume_20230215_SW_1900”
- [47] “Flume_20230215_GW_1” “Flume_20230215_GW_2”
- [49] “Flume_20231215_GW_4” “Flume_20230215_GW_6”
- [51] “Flume_20230215_FW_1” “Flume_20230215_FW_2”
- [53] “Flume_20230215_FS_1”

Export Processed Data

Export Processed Data

A tibble: 6 x 19

Project Site Replicate Sample_Type Sample_ID Date Time_EDT Time_EST
ic_mgL

1 SERC Lat~ Flume 0900 SW Flume_20~ 2/8/~ 9:00 10:00 10

2 SERC Lat~ Flume 1000 SW Flume_20~ 2/8/~ 10:00 11:00 9.72

3 SERC Lat~ Flume 1100 SW Flume_20~ 2/8/~ 11:00 12:00 6.94

4 SERC Lat~ Flume 1200 SW Flume_20~ 2/8/~ 12:00 13:00 9.50

5 SERC Lat~ Flume 1300 SW Flume_20~ 2/8/~ 13:00 14:00 9.46

6 SERC Lat~ Flume 1400 SW Flume_20~ 2/8/~ 14:00 15:00 9.90

i 10 more variables: ic_uM , ic_flag , Analysis_runtime ,

Run_notes , Field_notes , DO_mgL , pH ,

Salinity_ppt , Temp , Radium_Vol_L

““

#end